

Static Labor Supply

Labor I — Week 2

Teaching deck draft

Week 2 question

- How do workers choose whether and how much to work once taxes, transfers, and work costs matter?
- Why is the static model still the benchmark even though it is not the full story?
- Which policy debates are really debates about intensive versus extensive margins?

Why this week matters

- Week 1 gave us participation, hours, and earnings as measured objects.
- Week 2 gives the first worker-choice benchmark that maps tax wedges into those objects.
- Later weeks add dynamics, policy evaluation, and frictions, but they do not replace this benchmark.

Benchmark static problem

$$\max_{h \in [0, \bar{h}]} u(c, \bar{h} - h) \quad \text{s.t.} \quad c = y + wh - T(wh)$$

- Hours choice is already policy-facing because the schedule $T(\cdot)$ can be nonlinear.
- The model is narrow by design: wages and the schedule are taken as given.

$$\frac{u_\ell(c, \ell)}{u_c(c, \ell)} = w(1 - T'(wh))$$

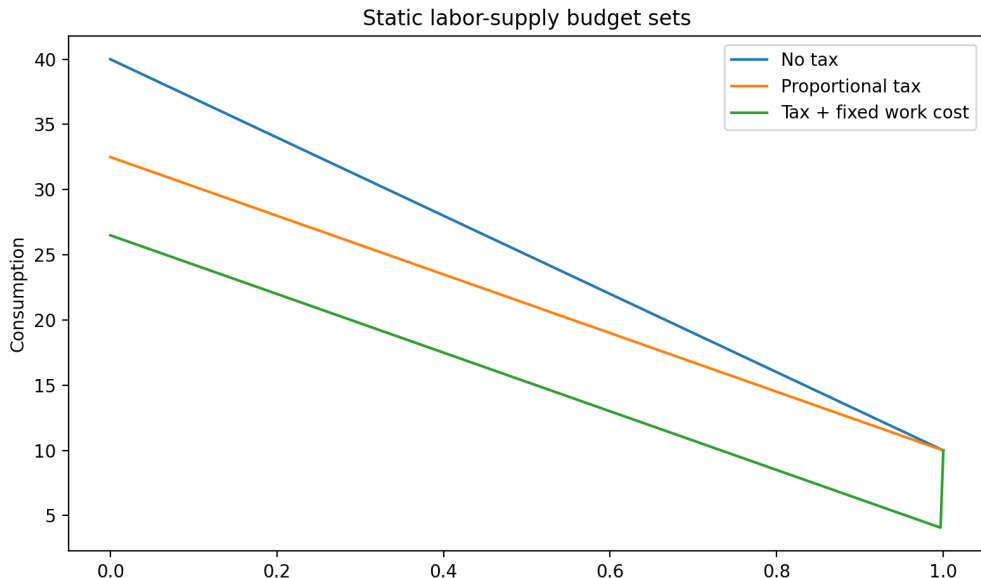
- The relevant price of leisure is the net-of-tax wage.
- Observed hours responses blend substitution effects, income effects, and heterogeneity.
- Small hours responses do not imply small participation responses.

Extensive versus intensive margins

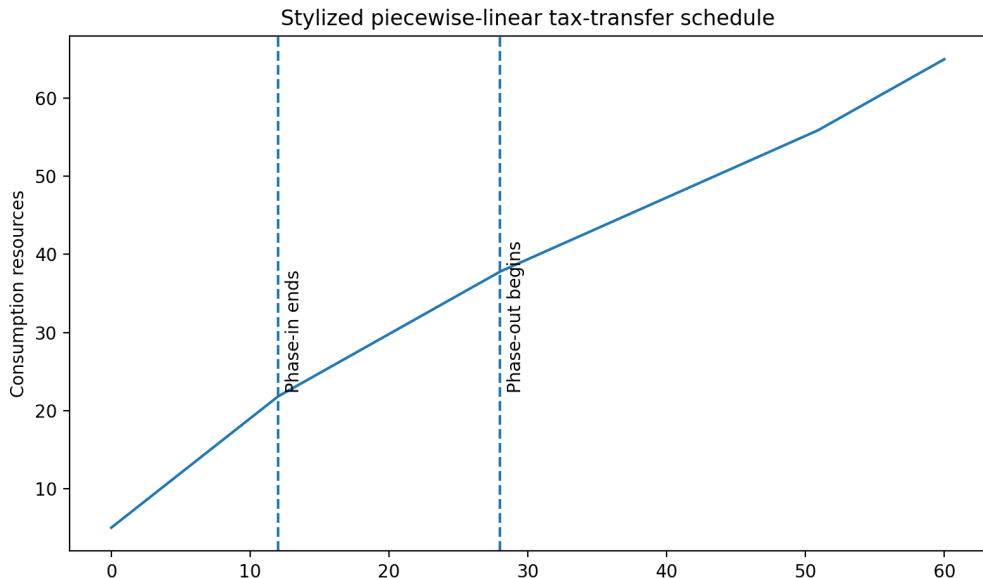
$$V^{\text{work}}(w, y, F) \geq V^{\text{nonwork}}(y)$$

- Intensive margin: hours conditional on work.
- Extensive margin: whether work clears the threshold once fixed costs F matter.
- Childcare, commute time, and benefit loss make the extensive margin first-order.

Budget sets and participation wedges



Nonlinear schedules are the policy norm



Elasticity taxonomy

Object		Margin	Main warning
Uncompensated elasticity	intensive	Hours	Mixes income and substitution effects
Compensated elasticity	intensive elasticity	Hours	Needs stronger structure
Participation elasticity		Work / nonwork	Sensitive to fixed costs and benefit loss
Taxable-income elasticity		Earnings / reporting	May not be real labor supply
Macro elasticity		Aggregate hours	Includes equilibrium and institutions

Why the EITC is the applied bridge

- Phase-in region raises the payoff to entering work.
- Plateau and phase-out regions create different marginal incentives for current workers.
- The same policy can move single mothers, married couples, and secondary earners differently.

- Eissa and Liebman: strong participation responses for single mothers.
- Eissa and Hoynes: married-couple responses depend on household earnings position.
- Saez: local bunching around kink points reveals responses to nonlinear schedules.
- Chetty, Friedman, and Saez: knowledge frictions shape observed EITC responses.

What Week 2 feeds next

- Week 3: dynamic labor supply and lifecycle adjustment.
- Week 11: worker-side policy evaluation across taxes, transfers, childcare, and leave.
- Week 12: information, attention, and behavioral frictions in labor supply.

- Static labor supply is the benchmark map from policy wedges to behavior.
- The key empirical question is often which margin is moving, not whether labor supply exists.
- Nonlinear schedules, fixed costs, and frictions make local interpretation essential.
- The EITC literature is the cleanest bridge from geometry to evidence.